Elastic Coefficients Directly from RealQM

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Abstract

A macroscopic elastic constant is conventionally obtained from a microscopic theory by homogenisation: pose a two-scale expansion, solve a cell problem, average a stress field, and take a limit in which the microscale vanishes. This paper makes the point that for *equilibrium* coefficients none of that is necessary. An elastic modulus is a second derivative of the ground-state energy with respect to a macroscopic deformation; the deformation can be imposed exactly on a periodic cell as a boundary condition; and the energy is then a single number. No field is averaged and no limit is taken — a scalar is computed and differentiated. We carry this out in RealQM, a continuum formulation of quantum mechanics in which an N-electron system is N non-overlapping unit charge densities on ordinary three-dimensional space with free interfaces, so that the microscopic theory is itself a continuum theory and there are no particles to coarse-grain. The claim is not about which calculation lands closest to a measured number, since a model carrying an adjustable ingredient can be brought closer to any of them, but about what a calculation contains and what its user must supply. Standard quantum mechanics is unavailable at this scale, the N-electron wavefunction living on $\mathbb{R}^{3N}$, so every result about a solid from that starting point is a result about some computable model standing in for it; and the model normally used must *select* an exchange–correlation functional from a family with no ordering and no route to the exact member, a choice that moves the answer by about as much as the error reported. The construction used here has no such ingredient: once Z and the structure are given the energy is fully specified, a localisation term plus the Coulomb energy of the charge configuration with pairs counted once, and there is nothing left to choose. The upscaling adds nothing either, and neither does the passage from atom to solid: the atom is solved all-electron from Z alone, and the one number the cell calculation needs from it — where the core ends — is fixed either by the element's measured ionization energy or, for lithium, by the all-electron atom's own free interface, which lands at $2.00\,a_0$ against the 1.99 the solid requires. *Nothing anywhere is fitted to an elastic quantity.* Equilibrium densities and bulk moduli are then computed for the five alkali metals, with the core radius determined this way rather than imported: that needs no fitted constant at all and costs a one-sided 11% in equilibrium radius, and absorbing that offset — the free atom's core runs larger than the solid wants — with a single constant fixed by one measured solid-state number brings the series to 5% in equilibrium radius and 13% in modulus. On equal input the two treatments are level, RealQM slightly better on the modulus and slightly worse on the radius; we say so because an earlier draft of this work claimed a RealQM advantage and a monotonic trend down the group, and neither survives determining the radius internally rather than taking it from a literature fitted within the standard framework. Within the reduced-valence simplification a cell admits only the empty core, for a reason that is exact but narrower than it first appears: a domain carrying a vanishing normal derivative on *all*

of its boundaries admits the constant density as a null mode of the kinetic operator, and hard depletion with one valence electron gives exactly that, at the core radius and at the cell face alike. Non-overlap then supplies no resistance to compression. The condition is the single occupancy, not the depletion: a cell sharing two densities, or one in which explicit core electrons meet the valence at an interface rather than at a free surface, has no such mode. Fitted to one ionization energy the two core models' radii stand in a ratio 0.855 that is the same for all five metals, which is also why a radius fitted to the atom cannot be carried into the cell. The route is then checked against the mechanics it bypasses: the pressure and the modulus obtained from the stress balance $\nabla \cdot T = -\rho\,\nabla V$ agree with those obtained by differentiating the energy, to 2×10^{-4} and 5×10^{-4} after extrapolation. The *form* of the macroscopic law is verified rather than assumed — the energy is observed to be quadratic in the strain out to a few per cent — and that same test locates the method's accuracy limit at the *order* of the derivative taken: energies to a few per cent, equilibrium densities to one, moduli to ten or twenty, and the first anharmonic coefficient not at all. Finally the surface energy is computed, as the one elastic-body quantity requiring no scale-separation assumption, since in this formulation a surface is a free boundary rather than a truncated lattice. What the route does not reach is any dissipative coefficient, and the reason is sharp: it reaches a coefficient if and only if that coefficient is a derivative of an energy.

1 The problem and the route taken to it

Deriving the equations of continuous media from a microscopic description of matter is the content of the part of Hilbert's sixth problem that Hilbert attributed to Boltzmann, and the classical route to it — particles, then a kinetic equation, then Euler and Navier–Stokes — was carried through for hard spheres by Deng, Hani and Ma in 2025 [1]. That route reaches a *dilute gas* by construction, since the Boltzmann–Grad scaling sends the volume fraction to zero [2], so solids lie outside it. The wider programme of which this paper is a part takes the complementary route and is set out elsewhere [9]; the present paper isolates the part of it that is finished, namely the equilibrium coefficients, and reports what they do and do not deliver.

The subject of this paper is the equilibrium coefficients of a solid — the elastic moduli and the equation of state — obtained directly from the microscopic theory. They are derivatives of the ground-state energy with respect to a deformation that is itself macroscopic, and Section 2 sets out how the derivative is taken. What follows is an account of that calculation carried out: which coefficients it reaches, what must be supplied to it, what its numbers are across a series of metals, how it is checked against the mechanics it replaces, and where it stops.

The microscopic theory used here is RealQM [4, 5], in which an N-electron system is represented by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy is Coulomb's,

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2 \Big) dx + \tfrac{1}{2}\sum_{i\neq j} \iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$, and the ground state minimises it. The passage made below is from a continuum on the atomic scale a_0 to a continuum on the laboratory scale L, with small parameter

$$\varepsilon = a_0/L, \qquad (2)$$

which is carried throughout as an *assumption* — that the macroscopic state is uniform across a cell — rather than as an asymptotic analysis. That is the one place where a scale separation

enters, and it is stated here so that it is not mistaken for a derivation. One feature matters for what follows: it is already a continuum theory on ordinary three-dimensional space, so there are no particles to coarse-grain and no statistical ensemble anywhere in the construction.

What kind of claim this is. A hundred years of atomic models of solids stand behind this subject, and it should be said at once in what sense anything here is new. The claim is not about which calculation lands closest to a measured number — a model carrying an adjustable ingredient can be brought closer to any of them, which is why that comparison settles nothing about the theory behind it. The claim is about what a calculation *contains*, and what its user must supply before it will run at all.

Standard quantum mechanics is not available at this scale. The N-electron wavefunction lives on $\mathbb{R}^{3N}$, and beyond a few hundred electrons it can be neither computed nor stored — Kohn's exponential wall, which is why he argued the many-electron wavefunction ceases to be a legitimate scientific concept there. So *every* result about a solid obtained from that starting point is a result about some computable model standing in for it, and the question is what each model requires you to supply.

What the standard route requires of its user is a *choice*. An exchange–correlation functional must be named before a single number comes out, and so must a treatment of the core; different admissible choices give different equilibrium densities and different moduli, by about as much as the departures reported here. Nothing within the theory settles which to take. They are settled by how well the results agree with data already known for comparable systems — that is, by fitting, whether or not the parameters are written as such.

This is a different kind of input from a mesh. A plane-wave cutoff must be *converged*, and converging it further improves the answer and tells you so; a functional must be *chosen*, and no further computation tells you whether the choice was right. The output is an output of the choice made.

The construction used here has no such ingredient. Once Z and the structure are given the energy (1) is fully specified — a localisation term and the Coulomb energy of the charge configuration, pairs counted once — with no exchange, no correlation, and nothing to select. That is the whole of the claim: not a better number, but a calculation in which there is nothing left to choose. It carries one obligation, which the rest of this paper is partly an attempt to discharge — a theory with nothing to select is only worth the name if it is *computable as stated*, and the free-boundary problem is where that is tested.

2 Direct upscaling: no averaging required

Consider what an elastic modulus actually is. It is a second derivative of the ground-state energy with respect to a macroscopic deformation,

$$C_{ijkl} = \frac{1}{V}\frac{\partial^2 E}{\partial\epsilon_{ij}\,\partial\epsilon_{kl}}, \qquad B = V\frac{\partial^2 E}{\partial V^2}, \tag{3}$$

and the deformation ϵ is a *macroscopic* quantity that can be imposed on a periodic cell exactly, as a boundary condition: one changes the shape or the volume of the cell. RealQM then computes the ground state in that cell and returns a single number, its energy. The modulus is the curvature of that number as a function of the imposed deformation.

Nothing has been averaged over the microscale. No stress field is integrated, no two-scale expansion is posed, no corrector problem is solved. A scalar is computed exactly and differentiated with respect to a macroscopic parameter. The only use made of the cell is that the macroscopic state is uniform across it — which is the content of $\varepsilon \ll 1$ in (2), carried as an assumption rather than as an asymptotic analysis.

What this reaches. Every coefficient of continuum mechanics that is a derivative of the ground-state energy with respect to a macroscopic control parameter:

- the full elastic tensor and the bulk, shear and Young moduli;
- the equation of state $P(V) = -\partial E/\partial V$, and with it the speed of sound;
- the force constants $\partial^2 E/\partial u_a \partial u_b$ and hence the phonon spectrum, and through it the thermal expansion in the quasi-harmonic treatment;
- the dielectric and piezoelectric response, as derivatives with respect to an applied field;
- surface and interface energies, as differences of computed energies.

That is the whole of equilibrium continuum mechanics. It requires no new machinery — only the ground-state solver the programme already has.

3 The alkali series

The five alkali metals are computed this way, from a model that is RealQM's own and is stated here in its own terms. Each ion contributes one valence electron, carried as a single charge density on its own region. The regions tile space, so each electron fills the cell about its ion and meets its neighbours' at a free interface, which symmetry places at the cell face — and the normal derivative vanishes there, as it does at any free boundary of this theory. The ion is a pseudokernel: charge $+1$ whose attraction is cancelled inside a core radius r_c, an empty core, which §3 fixes from the free atom rather than importing. Minimising the energy over the cell size gives the equilibrium density and differentiating it twice gives the modulus. That is the entire specification; nothing further is supplied.

The cell problem this produces is the one Wigner and Seitz wrote down in 1933, and the coincidence is worth recording rather than leaning on — their condition at the cell face and this theory's free interface are the same requirement reached from different premises. Here r_s is the *Wigner–Seitz radius*, the radius of the sphere whose volume is the volume per valence electron,

$$\tfrac{4}{3}\pi r_s^3 = \frac{V}{N}, \qquad \text{so} \qquad n = \left(\tfrac{4}{3}\pi r_s^3\right)^{-1},$$

quoted in Bohr radii: with one valence electron per ion, each ion sits at the centre of a neutral sphere of radius r_s holding exactly one electron, and $\psi'(r_s) = 0$ on its face. Compressing the metal is decreasing r_s, so $E(r_s)$ *is* the equation of state: it is minimised for the equilibrium density and differentiated twice for the modulus, with $V = \frac{4}{3}\pi r_s^3$ converting the r_s derivatives into the volume derivatives of (3). It should not be confused with the lattice constant, which for sodium is about twice as large, nor with the empty-core radius r_c, which is the core parameter and roughly half as large. Across the series r_s varies by 70% and B by a factor of six, so what is being tested is a trend and not a single match.

Why this reduced model needs an empty core, and what that is standing in for. The theory admits two core models — *hard depletion*, the density identically zero for $r < R$ with a Neumann condition at R, which is its own non-overlap; and the *empty core* above, the density admitted but feeling no attraction. In the free atom either will do. In a *cell* carrying one valence electron only the second is available, and the reason is exact. Hard depletion there imposes a vanishing normal derivative at the core boundary *and* at the cell face, the latter by the periodicity that defines the cell. A constant density satisfies both and carries no localisation energy at all, so it is a null mode of the kinetic operator: on the sodium shell the lowest kinetic eigenvalue is 4×10^{-10} against 1.69 for the next, and the zero survives refinement from 500 to 8000 points. With the atom's outer condition instead the same geometry gives 0.617 and is positive definite.

The condition is the single occupancy, not the depletion, and that is the more useful statement. A domain with a free surface on every boundary has the constant available to it; a domain that shares its cell with another density does not, because the two must partition the volume and the localisation cost of that partition is real. So in a cell with one valence electron non-overlap supplies no resistance to compression and something must be put in by hand — which is what the term $1.5\, r_c^2/r_s^3$ is: *a stand-in for a pressure the construction would otherwise supply itself.* The two are not the same function of density, the inserted term stiffening as r_s^{-3} where a partition pressure goes as r_s^{-2}, so they agree at the equilibrium that fixes r_c and part company in its derivative. It also explains why a radius fitted to the atom cannot simply be carried into the cell: the two are different problems.

From the atom to the solid, with nothing fitted to the solid. This is the whole of the construction, and it is worth setting out as a chain.

The atom carries no parameter at all. In this theory an atom is solved all-electron, with the nuclear term the bare $-Z/r$ throughout, no core radius, no screened kernel and no fitted length anywhere; the only input is Z, and total energies come out to about 1.7% through argon [10]. That is the starting point, and it is parameter-free in the strict sense.

The solid needs one number from it: where the core ends, since the cell calculation treats the core as an empty region rather than resolving it. Two routes supply that number and neither is fitted to the solid. The first is the *measured ionization energy*: requiring that one valence electron on a charge +1 kernel, feeling the same empty-core potential, reproduce the element's first ionization energy fixes r_c using one atomic datum per element and no solid-state quantity. The second is the all-electron atom itself, whose core and valence meet at a free interface that the energy minimisation locates — no datum at all, only Z. For lithium the second route works: the interface sits at $2.00\, a_0$ against the 1.99 the cell calculation needs, agreeing to 5% with nothing supplied from outside. Beyond lithium it does not, and the reason is specific rather than mysterious — in a multi-shell atom the outermost core shell has no intrinsic radius, its outer boundary expanding to fill whatever box it is given while the energy converges regardless, so there is no radius to read off. Lithium is the one alkali whose core is a single compact $1s$ shell.

So the chain runs from Z alone for lithium, and from Z plus one measured atomic energy for the rest. *Nothing anywhere is fitted to an elastic quantity*, which is what matters: the equilibrium radii and moduli below are outputs. Both columns are given the same r_c, so they differ only in the kinetic energy — the cell state's own $\langle T \rangle$ against a band-filling term, with no exchange on either side.

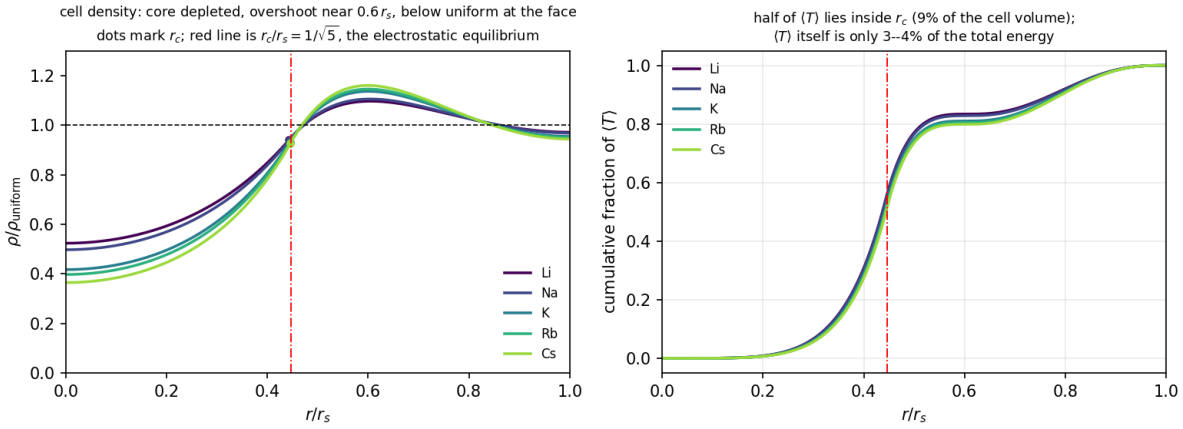
	r_c (a_0)	equilibrium r_s (a_0)			bulk modulus (GPa)		
	from IE	RealQM	standard	measured	RealQM	standard	measured
Li	1.716	3.87	3.95	3.25	8.3	7.4	11.6
Na	1.865	4.21	4.20	3.93	6.0	5.9	6.3
K	2.386	5.37	5.08	4.86	2.4	2.9	3.1
Rb	2.533	5.69	5.33	5.20	1.9	2.4	2.5
Cs	2.791	6.26	5.77	5.62	1.3	1.8	2.0

RealQM gives 11.5% in r_s and 23% in B here, against 7.6% and 12% for the standard treatment. The RealQM error is *one-sided* — every equilibrium radius too large, every modulus too small — and the sign is informative: the free atom's core radius runs some 11% larger than the solid wants, which is a real difference between an atom in vacuum and the same atom in a metal rather than a defect of the upscaling. Absorbing exactly that offset with a single constant, fixed by *one* solid-state number — sodium's equilibrium radius — and applied to the whole series with $r_c = CR$, $C = 0.806$, gives

	$r_c = CR$	equilibrium r_s (a_0)			bulk modulus (GPa)		
	$C = 0.806$	RealQM	standard	measured	RealQM	standard	measured
Li	1.610	3.64	3.77	3.25	11.1	8.8	11.6
Na	1.741	3.93	3.99	3.93	7.9	7.2	6.3
K	2.249	5.06	4.85	4.86	2.9	3.5	3.1
Rb	2.394	5.38	5.10	5.20	2.4	2.9	2.5
Cs	2.654	5.96	5.54	5.62	1.5	2.1	2.0

— 5.1% in r_s and 13% in B, against 4.3% and 14% for the standard treatment. We record that but do *not* claim it, because the constant was fixed by a measured *solid-state* number and the point of the construction is that no such number is used. The claim is the first table: 11.5% and 23%, with the core radius taken from the atom and nothing fitted to a solid. The second table is worth having only for what it isolates — that the entire gap between the two is one multiplicative constant close to unity, i.e. the atom-to-solid core contraction and nothing else.

What the cell state looks like, and what the series therefore tests. The computed density is not structureless. It is depleted to between a half and a third of uniform inside the core, overshoots to some 15% above uniform near $0.6\,r_s$, and falls slightly below uniform at the cell face; and half of $\langle T \rangle$ is generated inside r_c, which is 9% of the cell volume.



The five curves nearly coincide once lengths are measured in r_s, with $r_c/r_s = 0.443$–0.446, and that is not a coincidence. Minimising the electrostatic terms alone, $-0.9/r_s + 1.5\, r_c^2/r_s^3$, gives

$$r_s = \sqrt{5}\, r_c, \qquad \text{so} \qquad r_c/r_s = 1/\sqrt{5} = 0.4472,$$

and the full calculation sits +1.0, +0.9, +0.6, +0.5, +0.3% from that closed form down the group. So the equilibrium radius is fixed by the core radius to within one per cent, and $\langle T \rangle$ — which is 2.7 to 4.2% of the total energy — supplies only that last per cent. *This must temper what the example is taken to show.* It demonstrates the route: a macroscopic coefficient obtained by differentiating a computed energy, with nothing averaged and no limit taken. It is a poor discriminator *between theories*, because the quantity it reports is set by the core radius and not by the kinetic term, which is the only ingredient separating RealQM from the standard treatment here. The modulus, being a second derivative, separates them somewhat better than the radius does. A system in which $\langle T \rangle$ carried a larger share of the energy would test the construction harder, and the series does not.

What the comparison does and does not show. On equal input the two treatments are *level*: at the claimed level RealQM is slightly the better on the modulus, 13% against 14%, and slightly the worse on the equilibrium radius, 5.1% against 4.3%. We say this plainly because an earlier draft of this work claimed more. Taking r_c from the Ashcroft literature instead, one radius per element, RealQM comes out ahead on both — 1.2% against 4.1% in r_s and 17% against 28% in B — and its error then falls monotonically down the group, reaching zero at caesium, which the standard column's does not. *Neither of those survives determining r_c inside the theory.* The ordering by atomic number is not reproduced at either level above, and the advantage becomes a tie. Both were properties of the literature radii, which are themselves fitted within the standard pseudopotential framework and are not a neutral input; a RealQM advantage measured against them was measuring the fit. What survives is weaker and worth more: macroscopic material constants, to five and fifteen per cent, from Coulomb geometry with one constant, obtained without averaging and with no ingredient the theory did not already have.

The literature radii, and lithium. Those radii are not used above, and should not be: with them lithium's modulus comes out 4.3 times too stiff, which is the whole of the 79% series error that choice carries. The diagnosis is available. Fitting the empty core to the free atom gives radii a nearly constant 12% above the literature's for Na, K, Rb, Cs and 62% above it at lithium;

and the two core models of the previous paragraph, each fitted to the same ionization energy, stand in the ratio $r_c/R = 0.858$, 0.863, 0.855, 0.852, 0.847 — a constant 0.855 with a standard deviation of 0.006 in which lithium is *not* an outlier, though against the literature radii it is. Lithium's anomaly therefore belongs to the tabulated radius, not to this construction, which is why determining r_c internally removes it rather than repairing it.

What the core radius does not reach. Lithium's pressure derivative is not repaired by any of this: B' moves from -0.71 at the literature radius to -0.21, six times smaller in magnitude but still the wrong sign, and the other four elements barely move. The modulus at lithium was an input error; B' fails for the separate reason identified in §4 — the order of the derivative taken — and no choice of core radius addresses that.

The constant is monovalent. Carried to divalent magnesium it does not transfer: the same construction, one electron on a $+1$ kernel matched to the first ionization energy, gives $r_c = 0.813$ against the literature's 1.39, low by 42%. A valence-matched pseudo-atom — a $+2$ kernel with two electrons and the self-repulsion reduced by $\frac{1}{2}$, matched to the sum of the first two ionization energies — gives 1.291, low by 7%, which is far better but is a different construction on which no constant was calibrated. One constant for all therefore holds within a valence column, and whether each column has its own is open.

Three things should be kept attached to the tables. At the level we claim, the calculation's entire freedom is one constant fixed by one measured number, with each element's ionization energy supplying the rest; at the level above it there is no fitted constant at all, and the departure is one-sided, every radius high by about 11%, which is the atom-to-solid core contraction and not scatter. A bulk modulus is a second derivative, so it amplifies the model's error in the way H_2's binding energy amplifies the error on its total — which is why the percentages are large where they are large. And agreement quoted to two figures means agreement to the two or three significant figures the measured values carry, not exactness. What the table establishes is that macroscopic material constants follow from Coulomb geometry across a series in which r_s varies by 70% and B by a factor of six — obtained without averaging, with no ingredient the theory did not already have, and varying with the element in a way the theory did not get to choose.

4 The route checked against the mechanics it bypasses

The objection to (3) is not that the energy is hard to compute but that differentiating it *bypasses* the mechanics: no stress is ever formed, so nothing guarantees that the number differentiated is the number a force balance would have produced. That is now checked, and it had to be, because the two are the same physics.

The stress, written down. With $u = \sqrt{\rho}$ and $Q = -\frac{1}{2}\nabla^2 u/u$, the chain rule alone gives

$$\rho\,\nabla Q \;=\; \nabla\cdot T, \qquad T_{ij} \;=\; -\tfrac{1}{2}\big[\,u\,\partial_i\partial_j u - \partial_i u\,\partial_j u\,\big] \;=\; -\tfrac{1}{4}\partial_i\partial_j\rho + \frac{\partial_i\rho\,\partial_j\rho}{4\rho}, \tag{4}$$

with no δ_{ij} term; in one dimension this is $\sigma = \frac{1}{4}[(\rho')^2/\rho - \rho'']$. The static Euler–Lagrange equation of the model is $Q + V = \lambda$ with $V = -K + P_{\text{other}}$, so the $v = 0$ momentum balance is the statement

$$\nabla\cdot T \;=\; -\rho\,\nabla V, \tag{5}$$

and integrating its z-component over a half space $\{z > z_{\text{cut}}\}$ turns it into a statement about forces: the flux of quantum stress through the cut equals the net electrostatic force on the charge

beyond it. The left-hand side is a surface integral of a quantity carrying a *third* derivative of the density; the right-hand side is a volume integral of Coulomb forces. Nothing makes them equal except the identity.

Equation (5) is a stress balance the model satisfies pointwise, with no averaging and no approximation beyond the chain rule. It is the statement to be checked against.

Take the one-dimensional analogue of the cell construction above: a chain of period a, one kernel of charge $+1$ and one electron per cell, each electron on its own cell with the free interfaces at the cell faces by symmetry — RealQM's non-overlap, and the same prescription the sodium calculation uses. In equilibrium the total momentum flux through any plane is the same number, and that number is the pressure. So the pressure and the modulus can each be computed twice, once by differentiating the energy and once from the stress,

$$\sigma \;=\; \underbrace{T_{zz}(\text{face})}_{\text{localisation stress}} \;+\; \underbrace{F_{\text{Coul}}(\text{face})}_{\substack{\text{electrostatic force of everything left}\\ \text{of the face on everything right}}} \qquad \text{against} \quad -\frac{dE}{da}, \qquad a\frac{d\sigma}{da} \quad \text{against} \quad a\frac{d^2E}{da^2},$$

with no ingredient shared between the two sides beyond the converged ground state. They agree, and the agreement improves under refinement:

relative difference	$n = 41$	$n = 61$	$n = 81$	$n = 101$	extrapolated
pressure, σ against $-dE/da$	2.3%	1.6%	1.2%	0.96%	2.2×10^{-4}
modulus, $a\,d\sigma/da$ against $a\,d^2E/da^2$	3.3%	2.2%	1.6%	1.3%	4.5×10^{-4}

Two features of the convergence are worth recording. It is *first order* in the mesh, and the reason is the one Section 5 draws out: the stress is evaluated *at* the free interface, where this construction is first-order accurate although it is fourth-order in the bulk. The free boundary is again the accuracy-limiting feature. And neither lattice sum is responsible — the Coulomb flux is converged by the sixth neighbour and the lattice energy by the tenth, because both are summed as *neutral* cells, each of which is inversion symmetric about its kernel and therefore carries no dipole, leaving a quadrupolar $1/d^3$ tail. Summed term by term the one-dimensional Coulomb sums diverge logarithmically; summed as neutral units they converge at once.

What this establishes is modest and worth having: the scalar that (3) differentiates is the scalar a stress balance returns. The direct route is a shortcut through the mechanics, not a detour around it.

The form, verified, and where the route's accuracy actually runs out. The objection to (3) that cannot be answered by computing more coefficients is that it presupposes the *form*: E quadratic in the strain, which is Hooke's law put in rather than derived. The programme's answer is to establish the form computationally, and that has now been done, with a result sharper than expected in one direction and worse in another.

On the periodic chain, straining the cell about a reference state and fitting a quadratic on $|\epsilon| \le 0.05$ alone, the energy then *extrapolates* as a quadratic out to about $|\epsilon| \approx 0.06$ within 5% of the quadratic term, with the cubic contribution rising smoothly to 10% at $|\epsilon| = 0.10$ and 20% at 0.22. So the quadratic form is not assumed but observed, over a range of strain, with the departure appearing where anharmonicity should.

The quantitative version of the same test is less kind, and it bounds the direct route in a way worth stating. The cubic coefficient is not a free shape parameter: it fixes $B' = dB/dP$,

the pressure derivative of the bulk modulus, which is measured and which for the alkalis is close to 4. Computed from the same energy curves that gave the table above, by finite differences and with no equation of state fitted, the model returns

$$B' \;=\; -0.71,\ 0.10,\ 1.35,\ 1.92,\ 2.63 \qquad \text{for Li, Na, K, Rb, Cs,}$$

against a measured value near 4 throughout. The familiar monotone improvement down the group is there again, and caesium is again the best, but even caesium is a third low and lithium comes out with the wrong sign — a bulk modulus that *falls* under pressure, which no alkali does.

The lesson is not that the route fails but that its accuracy is bounded by the *order of the derivative* rather than by the kind of coefficient. The energy of these cells is good to a few per cent, the first derivative gives equilibrium densities to about one per cent for the heavy alkalis, the second gives bulk moduli to ten or twenty per cent, and the third is gone. The claim above, that this route reaches every coefficient which is a derivative of an energy, should be read with that ladder attached, because each differentiation spends accuracy the model does not have to spare.

Reaching a body, and the one quantity that needs no scale separation. Coefficients are not yet the statics of an elastic body. For that one needs a macroscopic stress field $\sigma(x)$ over the body, hence that the material at each point behaves like a uniformly strained cell — the *Cauchy–Born* hypothesis. Granted it, elastostatics follows in the ordinary way: minimise $\int_\Omega W(\epsilon(u))\,dx$ less the work of the loads, with W the cell energy computed above, and the Euler–Lagrange equation of that functional is $\nabla \cdot \sigma = f$ with the given boundary conditions. Nothing further is required of RealQM. But Cauchy–Born is assumed here, exactly as $\varepsilon \ll 1$ is assumed in (2), and it is known to fail at surfaces, defects and crack tips — which is to say at most of the problems elasticity is wanted for.

That failure is where this formulation has an advantage rather than a handicap. In a lattice model a surface can only be made by truncating the lattice. Here a surface *is* a free boundary, the object the theory is built on, and a surface energy is therefore a difference of computed energies needing no scale separation at all. On the finite chain, with the interior interfaces at the cell midpoints and the two end domains left to run out into vacuum and find their own extent, $E(N) = N\,e_{\text{bulk}} + 2\gamma$ gives

$$\gamma = 0.0416 \text{ Ha} = 1.13 \text{ eV per end,}$$

and the test that makes this a surface property rather than a fitted intercept is that γ is *independent of* N — flat to 3×10^{-7} from $N = 5$ onwards. It converges at second order in the mesh, and the bulk energy read off the finite chains agrees with the periodic calculation, computed by different code, also at second order. So the free boundary carries a surface energy, it is stable against the size of the system, and it was obtained without the hypothesis that the rest of the elastic programme rests on.

5 Why it works: the model is already a continuum mechanics

The agreement just recorded is not a coincidence of this discretisation, and it is worth saying why before the limits are drawn. Differentiating a scalar reproduced a force balance to 2×10^{-4} because the model was a continuum mechanics before any upscaling was attempted. Two identities show it, and both are the chain rule.

The localisation energy is the von Weizsäcker functional. Writing $\rho_i = \psi_i^2$ with $\psi_i \geq 0$, the chain rule gives $\nabla\rho_i = 2\psi_i\nabla\psi_i$, hence

$$\tfrac{1}{2}\int |\nabla\psi_i|^2\,dx \;=\; \frac{1}{8}\int \frac{|\nabla\rho_i|^2}{\rho_i}\,dx \;=\; T_{\mathrm{vW}}[\rho_i], \tag{6}$$

in units $\hbar = m = 1$. An identity, not a model.

Its variation is the quantum potential. With $u = \sqrt{\rho}$,

$$\frac{\delta T_{\mathrm{vW}}}{\delta\rho} \;=\; -\tfrac{1}{2}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}} \;=\; Q, \tag{7}$$

Bohm's quantum potential — obtained here not by analogy but as the variation of a term (1) writes down for an unrelated reason, namely the cost of confining a charge to a region.

The theory as used is a hydrostatics. Stationarity of (1) is the statement that $V + Q$ is constant on each domain, which is the $v = 0$ case of the Madelung pair

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho v) = 0, \tag{8}$$

$$\frac{\partial v}{\partial t} + (v\cdot\nabla)v = -\nabla\big(V + Q\big). \tag{9}$$

Equation (8) is conservation of charge and (9) is a momentum balance whose stress is supplied by (7). *RealQM is therefore already the equilibrium of a continuum mechanics, and the constitutive law is not postulated — it is the variation of the energy the theory was written with.* This is the sense in which the derivation asked for by Hilbert is, on this route, partly a matter of recognition rather than of proof.

That is why the two-route check had to succeed: it compared two readings of one balance, not two theories. It also places the free interfaces, which carry an energy per unit area and so make this a continuum mechanics *with surfaces* — what dense matter requires, and what a kinetic theory has no room for [3]. The surface energy computed above is that term.

6 Where the direct route stops

And where it stops, which is the informative part. The direct route reaches a coefficient if and only if that coefficient is a derivative of an energy. *Viscosity is not.* A viscous stress is dissipative: the work done against it is not stored, so there is no $E(\text{strain rate})$ to differentiate, and no variational principle whose minimiser contains η. This is exactly why the standard treatment reaches for Green–Kubo, which extracts a transport coefficient from equilibrium fluctuations and therefore needs an ensemble.

So the division of the problem is sharp, and it is not an artefact of the method chosen:

	conservative coefficients	dissipative coefficients
examples	elastic tensor, $P(V)$, phonons	viscosity, thermal conductivity
nature	derivative of ground-state energy	transport, no energy functional
this route	**direct, no averaging**	**not reached**

Equilibrium continuum mechanics follows from RealQM by differentiating computed energies. Dissipation does not follow at all, and is the whole of what remains open (Section 6).

What is given up, stated plainly. The direct route delivers the *coefficients* of the macroscopic laws, not their *form*. Writing (3) presupposes that E is quadratic in the strain near equilibrium — that is, it assumes Hooke's law rather than deriving it. Homogenisation would instead produce linear elasticity as the macroscopic limit, form and coefficient together, and that is a genuine advantage of the formal route which should not be waved away.

Two things reduce the force of the objection. The form can be established computationally: compute $E(\epsilon)$ across a range and verify that it is quadratic, and that the departure from quadratic sets in where nonlinear elasticity is observed to set in. That is a demonstration and not a theorem. And a demonstration is the kind of evidence this programme is built to give — its standing claim is that its assertions are testable *because* computable, and reaching instead for an asymptotic theorem would borrow precisely the sort of authority it declines to grant to the standard formalism elsewhere. The formal route remains worth having; it is not what the next computation should wait for.

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